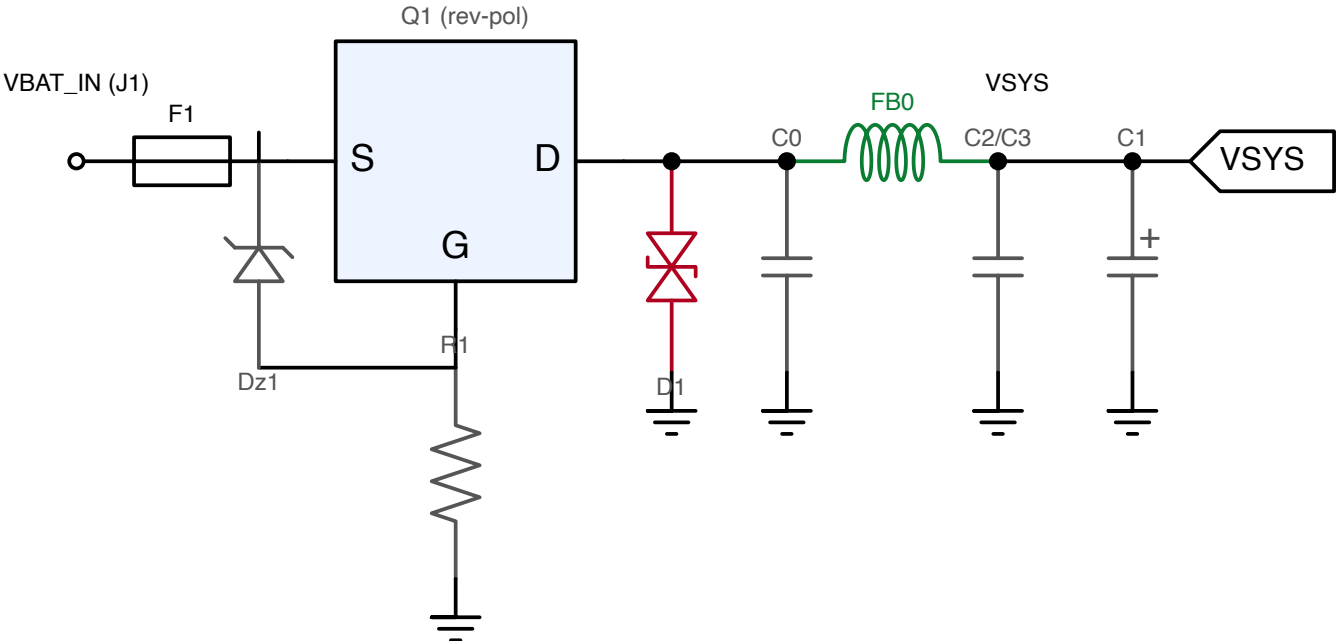
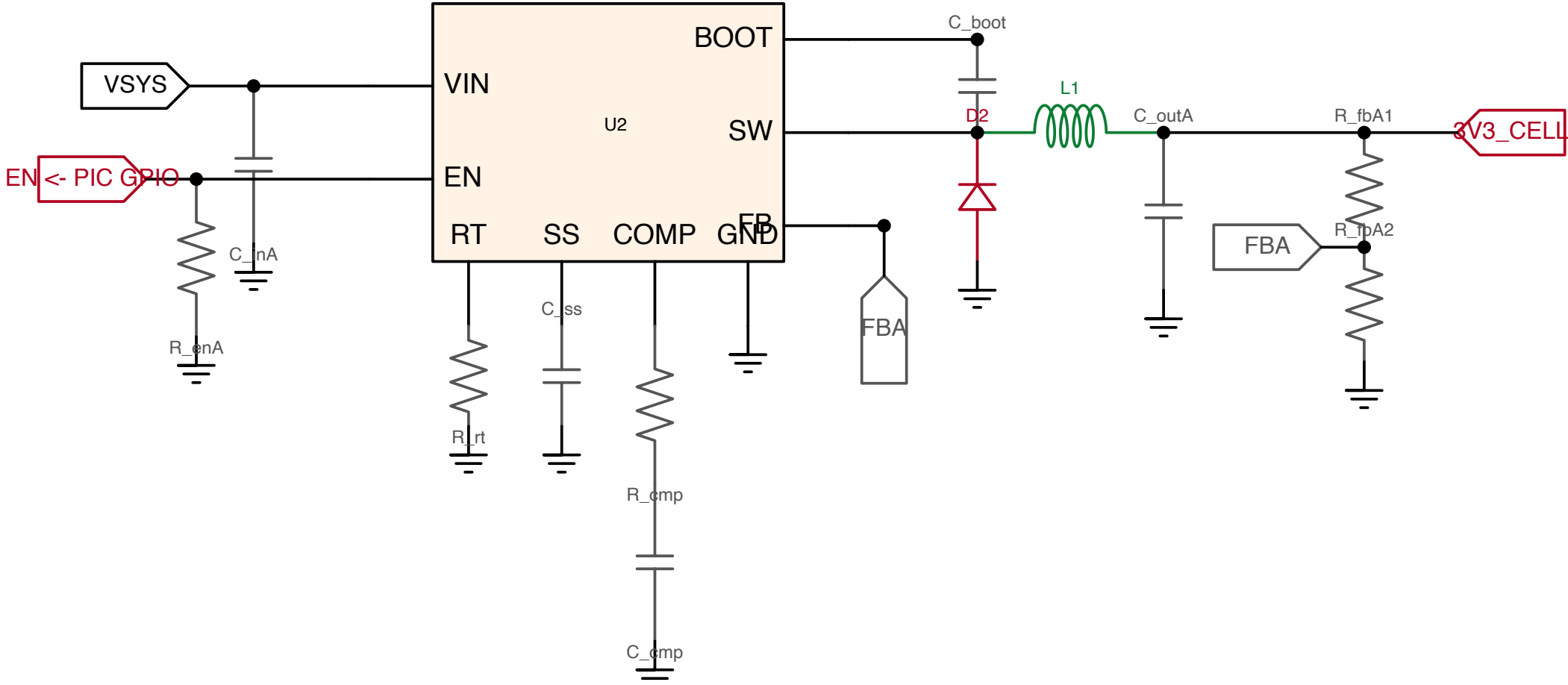


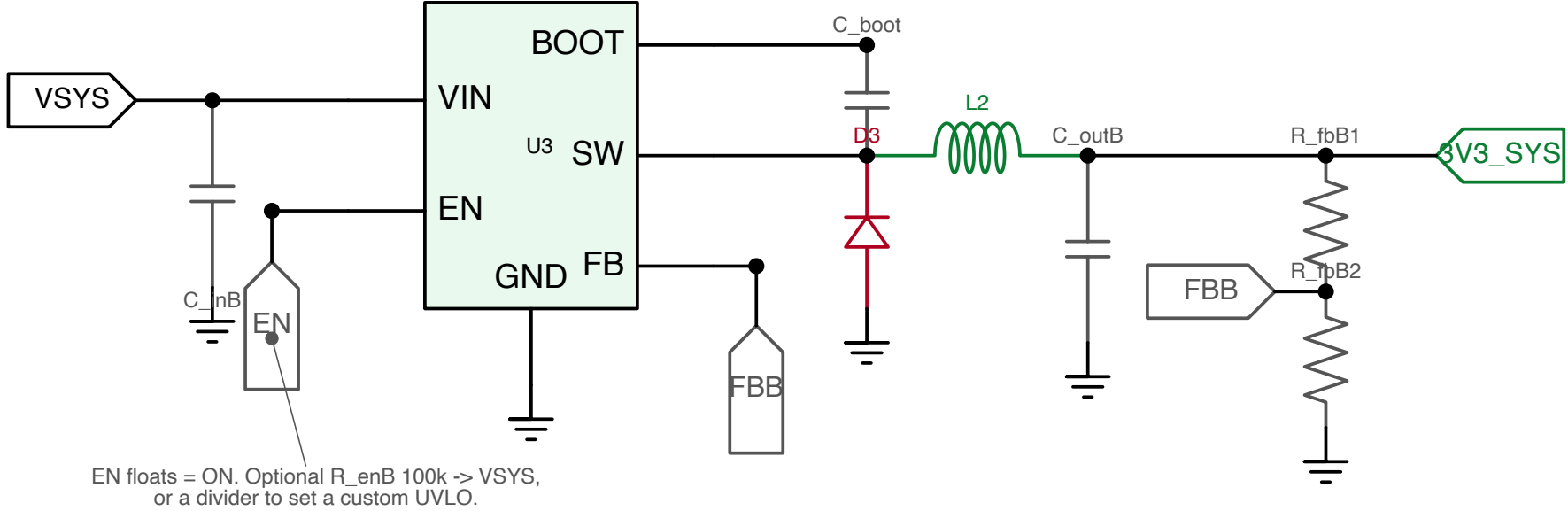
(1) VEHICLE INPUT -> PROTECTION -> VSYS



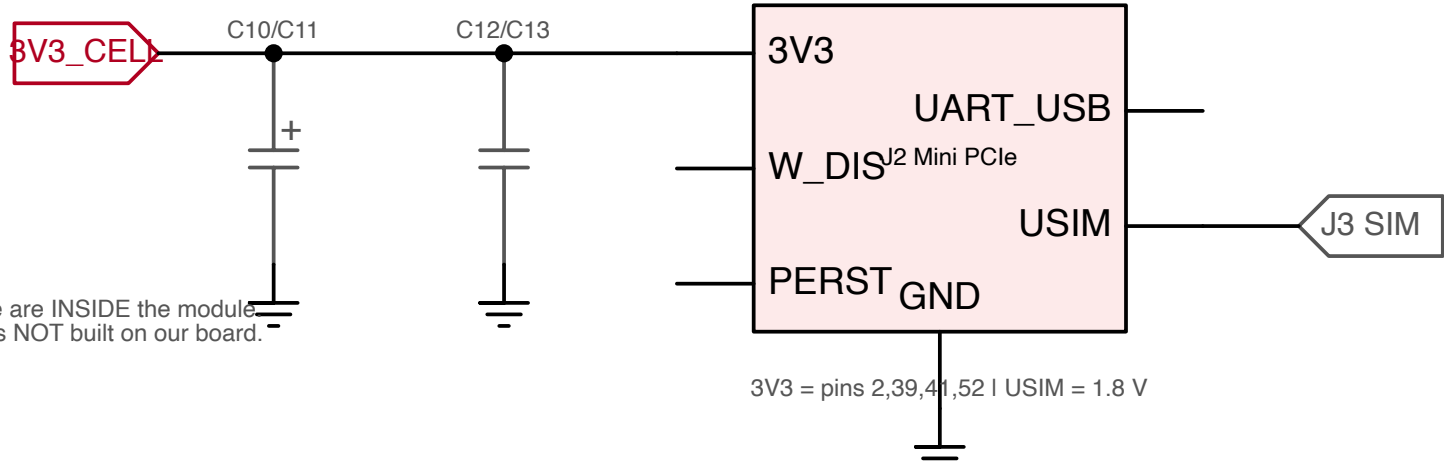
(2) BUCK A U2 TPS54560-Q1 : VSYS -> 3.3 V (≥ 4 A, 2G-capable) -> 3V3_CELL



(3) BUCK B U3 LMR16006X : VSYS -> 3.3 V (~0.6 A) -> 3V3_SYS (comp + soft-start internal)



(4) 3V3_CELL -> BG95-M3 MINI PCIe SOCKET



Card is self-contained: on-card VBAT regulation, RF/BB decoupling + ferrite are INSIDE the module.
The raw-module VBAT star network (ferrite + MLCC arrays + 2x TVS) is NOT built on our board.

3V3 = pins 2,39,41,52 | USIM = 1.8 V

(5) 3V3_SYS -> PIC VTG + NEO-M9N (ferrite-isolated GNSS branch)

